File permissions in Linux

Project description

This project is composed of different demonstration on appropriate Linux commands to manage file permissions on a Linux directory

Check file and directory details

The command to show and check for permissions is: `ls -l`

```
researcher2@cf9ae5be8cc2:~/projects$ ls -l
```

Describe the permissions string

Permission Analysis: Executing `ls -l` reveals the access control configurations for the directory contents. The `drafts` directory is restricted with owner read/write/execute and group execute permissions (`drwx--x---`), while the text files exhibit varying levels of access: `project_k.txt` is world-writable (`-rw-rw-rw-`), `project_m.txt` is restricted strictly to the owner and group (`-rw-r-----`), and both `project_r.txt` and `project_t.txt` grant write access to the group with read-only access for others (`-rw-rw-r--`).

```
total 20
drwx--x--- 2 researcher2 research_team 4096 Aug  6 04:53 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Aug  6 04:53 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Aug  6 04:53 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Aug  6 04:53 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Aug  6 04:53 project_t.txt
```

Change file permissions

Permission Remediation: Executing `chmod o-w project_k.txt` revokes write permissions for 'others' (world), hardening the file from an overly permissive state (`-rw-rw-rw-`) down to read-only for unprivileged users (`-rw-rw-r--`). The subsequent `ls -l` command verifies that world-write access was successfully stripped while maintaining standard owner and group access.

```
researcher2@cf9ae5be8cc2:~/projects$ chmod o-w project_k.txt
researcher2@cf9ae5be8cc2:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Aug  6 04:53 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Aug  6 04:53 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Aug  6 04:53 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Aug  6 04:53 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Aug  6 04:53 project_t.txt
```

Change file permissions on a hidden file

Hidden File Access Control: Executing `chmod u-w,g-w .project_x.txt` revokes write access for both the file owner and the group, converting the hidden file into a strict read-only state (`-r--------`). The subsequent `ls -la` command—using the `-a` flag to reveal hidden files—verifies that write permissions were successfully stripped for both entities, leaving only owner read access.

```
researcher2@cf9ae5be8cc2:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Aug  6 04:53 .
drwxr-xr-x 3 researcher2 research_team 4096 Aug  6 05:30 ..
-rw--w---- 1 researcher2 research_team   46 Aug  6 04:53 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Aug  6 04:53 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Aug  6 04:53 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Aug  6 04:53 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Aug  6 04:53 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Aug  6 04:53 project_t.txt
researcher2@cf9ae5be8cc2:~/projects$ chmod u-w,g-w .project_x.txt
researcher2@cf9ae5be8cc2:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Aug  6 04:53 .
drwxr-xr-x 3 researcher2 research_team 4096 Aug  6 05:30 ..
-r-------- 1 researcher2 research_team   46 Aug  6 04:53 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Aug  6 04:53 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Aug  6 04:53 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Aug  6 04:53 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Aug  6 04:53 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Aug  6 04:53 project_t.txt
```

Change directory permissions

Directory Access Restriction & Hardening: Executing the permission update strips all group and world permissions from the `drafts` directory, hardening its access state to `drwx------`. The resulting `ls -la` directory listing confirms that all access controls—read, write, and directory traversal (`x`)—have been completely revoked for both `research_team` group members and unprivileged external users, leaving the directory strictly isolated to the owner (`researcher2`).

```
researcher2@cf9ae5be8cc2:~/projects$ chmod g-x,o-x drafts
```

```
researcher2@cf9ae5be8cc2:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Aug  6 04:53 .
drwxr-xr-x 3 researcher2 research_team 4096 Aug  6 05:30 ..
-r-------- 1 researcher2 research_team   46 Aug  6 04:53 .project_x.txt
drwx------ 2 researcher2 research_team 4096 Aug  6 04:53 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Aug  6 04:53 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Aug  6 04:53 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Aug  6 04:53 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Aug  6 04:53 project_t.txt
```

Summary

Project Overview: This project provides a practical demonstration of managing and auditing file access controls across a Linux directory structure using terminal commands.
Auditing Access Controls (`ls -l` & `ls -la`): Used `ls -l` to analyze initial permission strings across standard files and `ls -la` to inspect hidden files (such as `.project_x.txt`) and directory traversal permissions (`.`).
Permission Remediation & Hardening (`chmod`):

- **Unprivileged Access Removal:** Applied `chmod o-w project_k.txt` to strip world-write access, securing the file down to a read-only state for external users (`-rw-rw-r--`).
- **Hidden File Isolation:** Executed `chmod u-w,g-w .project_x.txt` to revoke owner and group write permissions, locking the sensitive file into a strict read-only state (`-r--------`).
- **Directory Access Restriction:** Utilized `chmod` to remove group and world permissions on the `drafts` directory (`drwx------`), preventing unauthorized users from entering or inspecting its contents.

Verification Workflow: Followed every `chmod` permission modification with a relative `ls -l` or `ls -la` audit to visually confirm that the targeted permission bits were successfully updated and aligned with security policies.